Technology Stack

Date	30 SEPTEMBER 2026
Team ID	SWTID-2026-1660
Project Name	AI WEATHERWISE API
Maximum Marks	2 Marks

Define Your Technology Stack:

AI-Blog Nest API is built using Node.js and Express.js for backend development, with MongoDB and Mongoose for database management. Google Gemini AI is integrated to provide AI-powered features such as blog generation and intelligent assistance. JWT and bcrypt.js are used for secure authentication and password protection, while Helmet, CORS, dotenv, and Express Validator help improve security, configuration, and data validation.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML, CSS	React.js is used to build a responsive and interactive user interface. Its component-based architecture makes the UI reusable and easy to maintain. HTML and CSS are used for page structure and styling

2	Backend / Server-Side	Node.js, Express.js	Node.js provides fast and efficient server-side processing, while Express.js simplifies API development and routing. They are suitable for building scalable RESTful APIs for the AI BlogNest
3	Database / Data Storage	MongoDB	MongoDB is used to store blog posts, user information, comments, and other application data. Its flexible document-based structure is suitable for handling blog content and allows easy data management.
4	Cloud / Hosting / Deployment	Render / Vercel / MongoDB Atlas	Render can be used to deploy the Node.js and Express.js backend API, while Vercel can host the React.js frontend. MongoDB Atlas provides cloud-based database storage, making the application accessible, scalable, and easier to maintain.
5	Version Control & CI/CD	Git, GitHub, GitHub Actions	Git is used to track code changes and maintain different versions of the project. GitHub provides remote code hosting and collaboration. GitHub Actions automates testing, building, and deployment whenever new changes are pushed to the repository.

6	Third-Party APIs / Other Tools	Google Gemini API, Postman, Swagger/OpenAPI	Google Gemini API provides AI-powered blog generation and content assistance. Postman is used to test and debug REST APIs, while Swagger/OpenAPI is used to document and interact with the API endpoints.